INCH-POUND

MIL-S-43835E
15 June 1990
SUPERSEDING
MIL-S-43835D
30 March 1988

MILITARY SPECIFICATION

STRAPS, PACK FRAME AND FIELD PACK, GROUND TROOPS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers waist, shoulder, and chest straps for use by ground troops in attaching and adjusting LC-2 pack frames and field packs.

* 1.2 Classification. The straps or straps with pack frame shall be of the following items as specified (see 6.2).

Item 1 - Strap, waist, medium field pack, LC-2
Item 2 - Strap, waist, with lower back pad; pack frame, LC-2
Item 3 - Strap, shoulder, pack frame and field packs, right hand, LC-2
Item 4 - Strap, shoulder, pack frame and field packs, left hand, LC-2
Item 5 - Straps, waist and shoulder with pack frame, LC-2
Item 6 - Straps, chest for pack frame and field packs, LC-2

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
TT-E-529	-	Enamel, Alkyd, Semigloss, Low VOC Content
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-13924	-	Coating, Oxide, Black, for Ferrous Metals
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-F-43834	-	Frame, Field Pack, (Riveted), and Shelf, Cargo Support (Lightweight) LC-1

STANDARDS

FEDERAL

FED-STD-595	-	Colors
FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-2-312	-	Strap, Waist, Medium Field Pack, LC-2
2-2-416	-	Strap, Waist, with Lower Back Pad: Pack Frame, LC-2
2-2-417	-	Straps, Shoulder, Pack Frame and Field Pack, Left and Right Hand, LC-2
2-2-497	-	Strap, Chest
2-3-290	-	Frame, Pack, Ground Troops, Lightweight, LC-1; Assembly
2-3-493	-	Strap Packing Configuration
8-2-550	-	Clip, Retainer

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-5014.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel

D 1005 - Measurement of Dry-Film Thickness of Organic Coatings Using Micrometers

D 3951 - Standard Practice for Commercial Packaging

D 4181 - Standard Specification for Acetal (POW) Molding and Extrusion Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

* 3.2 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

* 3.3 Materials and components. Materials and components shall be as specified on the applicable drawings and as specified herein. For materials or components for which it is stated "or equal," if the contractor proposes to use an item considered to be equal to the material or component specified, prior to its use the contractor shall furnish a sample of the material or component, with supporting data to the contracting officer for subsequent evaluation by the responsible military agency. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

* 3.3.1 Fastener, steel. The steel fastener specified in Drawing 2-2-416 for the waist strap shall conform to Acorn Metal Products Co., Part Nos. C-103 and A-101; Brookside Manufacturing Co., Part No. 920; State Machine Products, Part No. SMP-1; State Recreation Inc., Part No. SR-1; ITW Waterbury Co., Part No. 917; or equal.

3.3.1.1 Finish, steel fastener. The rivets shall be given a black oxide coating conforming to class 2, 3, or 4, as applicable, of MIL-C-13924. The two lever components of the fastener shall be given a phosphate treated zinc plating conforming to type IV finish, service condition SC3 of ASTM B 633 followed by baked enameling. The baked enamel shall conform to type I or II of TT-E-529, color black. The coating on all components shall be applied prior to assembly and shall level out to a smooth and uniform film without orange peel, wrinkles, drops, streaks, or areas of no film (see 4.4.1.1).

3.3.1.2 Epoxy powder coating. As an alternate to the coating, zinc plating, and enameling specified in 3.3.1.1, the surfaces may be prepared and coated as follows:

a. The rivets and two lever components of the fastener shall be prepared for coating by mechanical means. The surfaces shall be thoroughly and uniformly textured by sand blasting with 80-grit aluminum oxide. Prepared surfaces shall be kept clean during

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subsequent handling. Handling to be by clean-gloved hands or by clean implements. The surfaces shall be retained in a clean environment. The surface shall be epoxy coated not more than 8 hours after sand blasting.

b. The finish coating shall be lusterless green (olive green) No. 34087 of FED-STD-595, electrodeposited epoxy powder coating conforming to H. B. Fuller Company, Kativo Powder No. KEP-43286-20 or equal.

c. The electrodeposited epoxy powder coating shall be baked in accordance with commercial practice.

d. The thickness of the baked coating shall be .003 ± .001 inch when tested as specified in ASTM D 1005.

e. The final coating shall be smooth and uniform in appearance, free from wrinkles and streaks (see 4.4.1.1).

* 3.3.2 Buckle, 2 inch, siderelease. The 2 inch plastic buckle specified on Drawing 2-2-416 for the waist strap shall conform to ITW Fastex Co., part No. SR-2D or equal. The material for the buckle shall be black Delrin. The buckle shall be rated at 250 pound tension (see 4.4.1.1).

3.3.2.1 Regrounds. Clean, unburned plastic material in the form of imperfect parts, sprues, runners, or other scraps of the same composition produced in the molding or finishing operation may be reground and mixed with the virgin material. When reground is mixed with the virgin material, the reground shall not exceed a level of 15 percent by weight of the blended mixture. The molding material may contain a surface lubricant.

3.3.3 Clip, retainer. The retainer clip for the quick release assembly on the shoulder strap shall conform to Drawing 8-2-550 and shall be made from virgin acetal molding compound conforming to ASTM D 4181 except that the electrical property requirements shall not apply. Clean, unburned plastic material in the form of imperfect parts, sprues, runners, or other scraps of the same composition, produced in the molding or finishing operation, may be reground and mixed with the virgin material. When reground is mixed with the virgin material, the reground shall not exceed a level of 15 percent by weight of the blended mixture. The molding material may contain a surface lubricant.

3.3.3.1 Color and finish. The color of the finished clip shall be black. The clip shall have a lusterless or dull finish on all surfaces. The finish shall be produced as a result of the molding operation and not by application of any protective coating, lacquer, or other materials. The parts shall be clean, smooth, and free from cracks, dirt, sink marks, and flash. All gates shall be trimmed flush with the molding surfaces.

3.3.4 Frame, pack (for item 5). The pack frame, when required, shall conform to MIL-F-43834 and shall be furnished without the cargo support shelf.

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3.4 Construction. The construction of the straps shall conform in all respects to the drawings listed in section 2 and as specified herein.

3.4.1 Stitching. All stitches, except bartacking, shall conform to stitch type 301 of FED-STD-751. The stitches per inch shall be as follows:

For all stitching except stitching to be sewn through foam pad	8-10 stitches per inch
For stitching required to be stitched through foam pad	6-8 stitches per inch

3.4.1.1 Type 301 stitching. Ends of stitching shall be backstitched or overstitched not less than 1 inch (1/2 inch for box-X and W-W stitching) except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose bobbin or top thread or excessively tight stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The inter lock shall be embedded in the materials sewn.

3.4.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch (1/2 inch for box-X and W-W stitching) back of the end of the stitching.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area (1/2 inch for box-X and W-W stitching), continue over the defective area, and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. When damaged thread has been removed, needle holes do not constitute damage to the material.

(When making the above repairs in a. and b. above, the ends of the stitching are not required to be backstitched.)

3.4.1.2 Bartacks. Unless otherwise specified, bartacks shall be as specified on the applicable drawing and as follows:

Length	Width	Stitches per bartack
1/2 inch $\pm$ 1/16	1/8 inch $\pm$ 1/32	28
3/4 inch $\pm$ 1/16	1/8 inch $\pm$ 1/32	42

Bartacks shall be free from thread breaks and loose stitching.

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3.4.1.3 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three or more tying, overlapping, or back stitches are used to secure the ends of stitching.

3.4.1.4 Thread ends. All thread ends that are visible on the finished item shall be trimmed to a length of not more than 1/4 inch.

3.4.2 Setting of snap fasteners. A hole shall be prepunched to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials and no more than three splits shall occur in the button or eyelet barrels.

3.4.3 Setting of eyelets. Holes shall be prepunched to receive the eyelets. The hole shall not be punched in the setting operation with the eyelet barrel. Holes prepunched to receive the eyelets shall be smaller than the outside diameter of the eyelet barrel so that the barrel must be forced through the hole. The eyelet shall be securely clinched without splitting and in a manner that will securely retain the loop fasteners.

3.4.4 Fusing of ends of nylon cord and webbing. The ends of the nylon cord and webbing shall be fused. The apparatus used to fuse the webbing ends shall be capable of providing sufficient heat to provide a smooth edge and with the cut ends of the webbing yarns all fused together. Fusing of the webbing ends shall be accomplished prior to the stitching operation.

3.5 Identification marking. Each strap shall be marked, in the location shown on Drawing 2-2-312, 2-2-416, or 2-2-417, as applicable, in accordance with type IV, class 5 of DDD-L-20.

3.6 Assembly of straps on pack frame. Item 5 (see 1.2) shall consist of the straps identified as items 2, 3, 4, and 6, and the pack frame conforming to 3.3.4. The straps shall be assembled to the pack frame as follows:

(NOTE: Reference to the left or right side of the pack frame shall be as viewed when viewing the user from the rear).

a. Waist strap. The waist strap, with back pad, shall be assembled to the pack frame with the male end of the plastic buckle on the left side of the frame. Pass the ends of the waist pad attaching strap (item 1, Drawing 2-2-416) through the loops of the pack frame (item 17, Drawing 2-3-290). Thread the frame attaching strap (item 7, Drawing 2-2-416) through the loops of the waist pad attaching strap then thread the free end of the strap through the fastener as shown in section A-A, Drawing 2-2-416 and tighten securely.

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b. Shoulder straps, typical left and right. Install left and right shoulder straps on the left and right sides of the frame as follows:

1. Insert loop of adjustment strap through plastic grommet in bottom cross member of frame and pass opposite end through loop (looped end of strap shall be on the outside surface of cross member). Tighten to a snug fit.

2. Insert frame attaching strap through web guide (item 32, Drawing 2-3-290) continuing up and over top of frame body and then through buckle on shoulder strap. Tighten to a snug fit.

3. Connect adjusting strap to upper portion of shoulder strap by securing the quick release assembly as shown on Drawing 2-2-417.

c. Chest strap. Assemble the chest strap onto the shoulder straps by pushing the left and right shoulder straps through the appropriate loops of webbing of the chest straps.

3.7 Repairs. Repairs such as mends, darns, or patches are not permitted on the straps.

3.8 Replacement of defective components. During the spreading, cutting, and manufacturing process, components of the strap having material defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.

* 3.9 Workmanship. The finished straps and straps with pack frame shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. Any nonconformance shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

* 4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the following conform to the specified requirements:

a. Foam pad material (see Drawings 2-2-416 and 2-2-417).
b. Finish on steel fastener (see 3.3.1.1).
c. Coating and thickness (when applicable) on steel fastener (see 3.3.1.2).

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d. Material for belt buckle (see 3.3.2).
e. Dimensional requirements of the chest strap, belt buckle, and retainer clip covered by Drawings 2-2-497 and 8-2-550.

* 4.4.2 In-process inspection. Inspection shall be made of the following operations to establish conformance to specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

a. Dimensional examination of parts for which a cut length or size is specified to determine conformance with specified dimensions.

b. Visual examination of cut parts to remove parts containing material defects and damages (see 3.8).

c. Visual examination of subassemblies to determine if prepunched holes for receiving snap fasteners are in conformance with 3.4.2. Examination of construction details which cannot be examined in the end item.

d. Visual examination for fusing of ends of webbing which are not easily visible in the assembled item.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of straps. The sample unit shall be one strap. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10 for total (major and minor combined) defects.

* TABLE I. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Fabric	Hole, cut, tear, smash, broken or missing yarn, or open place clearly visible at normal inspection distance (approximately 3 feet)	101	
	Shade bar or abrasion mark		201
Cord	End not fused		202
	Any cut	102	
	Frayed or abraided		203

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* TABLE I. End item visual defects - Continued

Examine	Defect	Classification	
Webbing	Any hole, cut, tear, or smash	103	
	Not firmly and tightly woven, or edges frayed or scalloped	104	
	Multiple float		204
	Abrasion mark, slub, broken end, or pick		205
	Cut end not fused as specified	105	
Hardware, general	Broken or malformed, failing to serve intended purpose, corroded area, burr, or sharp edge	106	
	Any required component improperly installed causing failure to serve intended purpose	107	
	Size or type not as specified	108	
Finish	Missing or not as specified	109	
	Color of finish is not as specified		206
	Coating is not smooth or uniform in appearance		207
	Evidence of sag, run, pinholes, blister, wrinkles, frothing or streak		208
	Area of no coating		209
	Exposed bare metal due to scratch, scuff, dig, or abrasion		210
Belt buckle and retainer clip (plastic)	Color not as specified		211
	Color not molded in		212
	Broken, cracked, or chipped	110	
	Any sink mark		213
	Any flash	111	
	Gates not trimmed as specified		214
	Surface not smooth		215
Snap fasteners	Any fastener not functioning properly, i.e., fails to snap closed, provide a secure closure or to open freely	112	
	NOTE: The fasteners shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.		

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* TABLE I. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Snap fasteners (cont'd)	Clinched excessively tight, cutting adjacent material	113	
	Clinched loosely, permitting any component to rotate freely but not to the degree that any component can be expected to become detached during use		216
	Clinched loosely to the degree that components can be expected to become detached during use	114	
	Incorrect style	115	
	More than three splits in eyelet or button barrels		217
Eyelets (for securing loop fasteners)	Insecurely clinched	116	
Open seams	More than 1/2 inch	117	
	1/2 inch or less		218
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or runoff stitches occur.		
Raw edge (on edge required to be finished)	More than 1/2 inch when securely caught in stitching		219
NOTE:	Raw edge not securely caught in stitching shall be classified as an open seam.		
Run-off (see open seam)			
Seam and stitch type	Wrong seam or stitch type	118	

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* TABLE I. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Stitch tension	Loose, resulting in loose top or bobbin thread		220
	Excessively tight resulting in puckering of material		221
Stitches per inch	One stitch less than minimum specified		222
	Two or more stitches less than minimum specified	119	
	Two or more stitches in excess of maximum specified		223
	NOTE: Variation in the number of stitches per inch caused by the operator speeding up the machine and pulling the material in order to sew over heavy places or heavy seams, or turning corners shall be classified as follows: (a) Within the minor defect classification - no defect. (b) Within the major defect classification - minor defect.		
Thread breaks, skipped stitches, or runoffs	Not overstitched as specified		224
	NOTE: Thread breaks or two or more consecutive skipped or runoff stitches not overstitched shall be classified as open seams.		
Ends of stitching	Not secured as specified		225
Rows of stitching	Any row missing except on W-W stitching	120	
	On W-W stitching:		
	- One row of stitching omitted		226
	- Two or more rows of stitching omitted	121	

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* TABLE I. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Bartacks	Loose stitching, incomplete, or broken		227
	Any omitted	122	
Components and assembly	Any component or required operation omitted (unless otherwise classified herein)	123	
	Any component misplaced or not assembled as specified	124	
	Needle chew		228
	Ends of straps not cut on a bias as specified		229
	Any mend, darn, or patch	125	
	Ends of release cord not secured as specified or not securely tied		230
Shoulder strap and waist strap (Drawing 2-2-312 and 2-2-417) quick release assembly	Improperly constructed preventing proper assembly or causing failure to function when released	126	
	NOTE: To test the function of the quick release assemblies, the components shall be assembled and released by pulling on the end of the release strap.		
Cleanness	Grease or oil stains clearly noticeable		231
	Thread ends not trimmed to 1/4 inch maximum length		232
Identification markings	Omitted, incorrect, illegible, or misplaced		233

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of straps. The sample unit shall be one strap. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.5 End item assembly of straps on packframe examination (item 5 only). Item 5 end items assembled from pack frames and straps which have previously passed the examinations specified in MIL-F-43834, 4.4.3, and 4.4.4 shall be examined for the defects listed below. The lot size shall be expressed in units of pack frames with attached straps. The sample unit shall be one pack frame with attached straps. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Waist strap	Not attached to pack frame with male end of belt buckle on left side of frame Not secured to pack frame as specified
Chest strap	Not attached to shoulder straps
Shoulder straps	Not attached on proper side of pack frame, i.e., right strap attached to left side or vice versa Adjusting strap or frame attaching strap not attached to pack frame as specified Quick release assembly not assembled

* 4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

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4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation.

5.1.1.1 Straps (when procured separately). Each strap shall be neatly and compactly folded, and then secured at each end with tape or twine, except that the medium field pack waist strap shall be secured once around the center.

* 5.1.2 Commercial preservation. Straps shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing.

* 5.2.1.1 Strap and pack frame assembly. Four strap and pack frame assemblies reverse nested and separated by a fiberboard divider pad as shown in drawing 2-3-493, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class weather resistant, variety DW, grade V15c of PPP-B-636. Inside dimensions of the box shall be approximatlely 23-1/2 inches in length, 13 1/2 inches in width, and 11 inches in depth. Approximate dimensions are furnished as a guide only. The divider pad shall be the full length and width of the box and constructed from corrugated fiberboard. Each container shall be closed in accordance with

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method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.1.2 Straps (when procured separately). Two hundred medium field pack waist straps or thirty-five pack frame waist straps with lower back pad or one hundred and twenty right hand pack frame and field pack shoulder straps, or one hundred and twenty left hand pack frame and field pack shoulder straps or two hundred chest straps, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing.

* 5.2.2.1 Strap and pack frame assembly. Four strap and pack frame assemblies reverse nested and separated by a fiberboard divider pad as shown in drawing 2-3-493, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Inside dimensions of the box shall be approximatlely 23-1/2 inches in length, 13-1/2 inches in width, and 11 inches in depth. Approximate dimensions are furnished as a guide only. The pad shall be the full length and width of the box and constructed from corrugated fiberboard. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

* 5.2.2.2 Straps (when procured separately). Two hundred medium field pack waist straps or thirty-five pack frame waist straps with lower back pad or one hundred and twenty right hand pack frame and field pack shoulder straps, or one hundred and twenty left hand pack frame and field pack shoulder straps or two hundred chest straps, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each container shall

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have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.3 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Straps only, preserved as specified in 5.1, or strap and pack frame assembly shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), strap and pack frame assemblies only or straps only, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 91 in accordance with appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The waist strap identified as item 1 in 1.2 is for use with the medium field pack. The waist strap identified as item 2 is for use with the pack frame. The shoulder straps identified as items 3 and 4 are for use with the pack frame or the medium field pack. The straps, waist and shoulder with pack frame identified as item 5 are used to carry medium or large field packs, or cargo. The chest strap identified as item 6 is for use with the shoulder straps on medium field pack and pack frame.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Item required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.3).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Edge fraying prevention. Cut edges of nylon cloth are subject to fraying during the process of handling the cut parts. It has been found that fusing of the cut edges of the cloth will prevent fraying.

6.6 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG No. 2311. When amendment, revision, or cancellation of this document is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.7 Subject term (key word) listing.

Chest strap
LC-2
Shoulder strap
Waist strap

6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - NU
Air Force - 99

Review activities:

Army - MD
Navy - MC
Air Force - 82
DLA - CT

User activities:

Navy - YD
Air Force - 45

Preparing activity:

Army - GL

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